

Pre-Allocated, Not Prevented

Loaner-Laptop Turn-Aways, Forecast-Fed Versus Ad Hoc, During Exam Week

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The problem

Week-ahead demand forecasting is meant to route loaner laptops to where exam-week demand will peak. Little evidence covers what happens when only one branch of a shared collection gets it: does pre-allocation cut turn-aways, or just relocate the queue?

Research questions

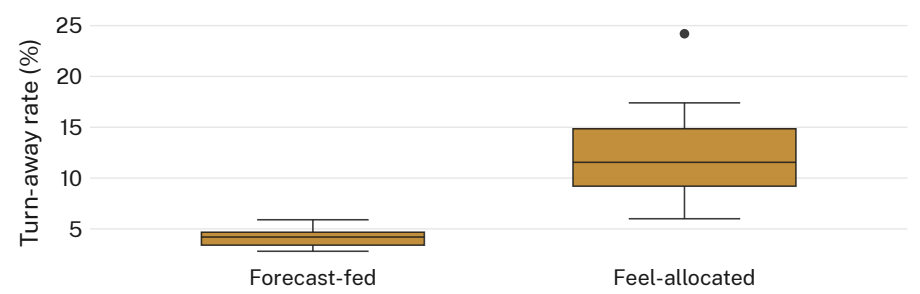
- Compare exam-week turn-away rates between a forecast-fed branch and its feel-allocated sister branch.
- Test whether forecast-driven pre-allocation reduces turn-aways or displaces demand onto the neighbouring desk.
- Check whether the non-forecast branch's unmet demand tracks the forecast branch's surplus.

Study design

- 2 branches · kiosk sign-out + waitlist logs · 5-min polling · 8-day exam period (n = 214 sessions, 61 turn-aways)
- Week-ahead demand forecast (rolling 7-day loan history, queue sensor) drove pre-allocation at the treatment branch only
- Sister branch kept allocating by staff judgement after its forecasting rollout stalled at integration; no other service difference held constant

What we found

Median exam-week turn-away rate held to 4.2% at the forecast-fed branch (range 2.8-5.9%) against 11.6% at the feel-allocated branch (range 6.0-24.2%), a 2.8× gap. Aggregate loaner utilisation across both branches was comparable, consistent with relocation rather than net reduction of unmet demand.



Daily turn-away rate distribution, 8-day exam period, 2–9 June 2026: the feel-allocated branch's median (11.6%) runs 2.8× the forecast-fed branch's (4.2%), with a single day reaching 24.2%.

Implications

Pre-allocation flattens the demand curve at the branch it serves; aggregate turn-away across both branches barely moved, consistent with relocation rather than reduction. Whether displaced students present at the sister desk or forgo a loaner is unresolved from sign-out logs alone. Next: match borrower IDs across branches to separate displacement from foregone demand.

Selected references

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De-identified sign-out logs only; no borrower ID linked across branches for this analysis.

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